An overpartition congruence modulo 7: proof of a conjecture of Wei

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Abstract

Let $\overline{p}(n)$ denote the number of overpartitions of n. Wei conjectured that $\overline{p}(16n) \equiv 0 \pmod 7$ whenever $n \equiv 3 \pmod 8$ and $\left(\frac{n}{7}\right) = 1$. We prove this. Theta-function 2-dissections give an exact expression for $\sum \overline{p}(16n)q^n$; reduction modulo 7 turns it into a holomorphic modular form of weight 18 on $\Gamma_0(224)$ divided by a power series in q^{56}; a twisting argument and Sturm's theorem finish the proof.

1 Introduction

Overpartitions have generating function

$$\sum_{n\geq 0} \overline{p}(n)q^n = \prod_{k\geq 1} \frac{1+q^k}{1-q^k} = \frac{1}{\varphi(-q)},$$

where $\varphi(q) = \sum_{k\in\mathbb{Z}} q^{k^2}$. Wei [5] proved Ramanujan-type congruences $\overline{p}(11(8n+5)) \equiv 0 \pmod{11}$ and $\overline{p}(13\cdot 2^6(8n+7)) \equiv 0 \pmod{13}$ and stated:

Conjecture ([5, Conjecture 4.1]). If $n \equiv 3 \pmod 8$ and $\left(\frac{n}{7}\right) = 1$, then $\overline{p}(2^4 n) \equiv 0 \pmod 7$. Equivalently, $\overline{p}(16(56n+k)) \equiv 0 \pmod 7$ for all $n \geq 0$ and $k \in \{11, 43, 51\}$.

Theorem 1. *For every integer $n \geq 0$ and $k \in \{11, 43, 51\}$,*

$$\overline{p}(16(56n+k)) \equiv 0 \pmod 7.$$

Equivalently, $\overline{p}(16m) \equiv 0 \pmod 7$ for every $m \geq 0$ with $m \equiv 3 \pmod 8$ and $\left(\frac{m}{7}\right) = 1$.

Ryan et al. [3, Theorems 1.3 and 1.5] already give square-prime, Legendre-restricted families, as well as cube-prime families. Our distinction is the actual target: the arguments here are prime to 7, whereas their displayed modulo-7 families contain that prime as a factor. We claim this specialised congruence, not a new general twisting method. Notation: $q = e^{2\pi i z}$, $\operatorname{Im} z > 0$; $E[f](q) = \sum_n f_{2n} q^n$ for $f = \sum f_n q^n$, so $E[f](q^2) = \frac{1}{2}(f(q) + f(-q))$; and $S_k(q) = \sum_n \overline{p}(kn)q^n$. Negative-index coefficients are zero.

2 Dissections

We use three classical identities (Berndt [1, Ch. 16, Entry 25]):

(a) $\varphi(q) + \varphi(-q) = 2\varphi(q^4)$, (b) $\varphi(q)\varphi(-q) = \varphi(-q^2)^2$, (c) $\varphi(q)^2 + \varphi(-q)^2 = 2\varphi(q^2)^2$.

Lemma 2.

$$S_2 = \frac{\varphi(q^2)}{\varphi(-q)^2}, \quad S_4 = \frac{\varphi(q)^3}{\varphi(-q)^4}, \quad S_8 = \frac{E[\varphi^7](q)}{\varphi(-q)^8}, \quad S_{16} = \frac{E\big[E[\varphi^7]\,\varphi^8\big](q)}{\varphi(-q)^{16}}.$$

Proof. By (b), $1/\varphi(-q) = \varphi(q)/\varphi(-q^2)^2$, and $E[\varphi](q) = \varphi(q^2)$ by (a), giving S_2. By (b) again, $S_2 = \varphi(q^2)\varphi(q)^2/\varphi(-q^2)^4$, and $E[\varphi^2] = \varphi^2$ by (c), giving S_4. Writing $S_4 = \varphi(q)^7/\varphi(-q^2)^8$ (by (b)) gives S_8, and writing $1/\varphi(-q)^8 = \varphi(q)^8/\varphi(-q^2)^{16}$ gives S_{16}. □

3 Reduction modulo 7

Reduction modulo 7 is a ring homomorphism $\mathbb{Z}[[q]] \to \mathbb{F}_7[[q]]$ commuting with E and with inversion of series with constant term 1, and $f(q)^7 \equiv f(q^7)$.

Lemma 3. *With $X = \varphi(q)^4$, $Y = \varphi(-q)^4$: $E[\varphi^8] = 8X^2 - 8XY + Y^2$.*

Proof. Put $a = \varphi(t)^2$, $b = \varphi(-t)^2$ with $t^2 = q$. By (c) and (b), $a + b = 2\varphi(q)^2$ and $ab = \varphi(-q)^4$, so $a^2 + b^2 = 4X - 2Y$ and $a^4 + b^4 = (4X - 2Y)^2 - 2Y^2$. Then $E[\varphi^8](q) = \frac{1}{2}(a^4 + b^4)$. □

Since $\varphi^7 \equiv \varphi(q^7)$, $E[\varphi^7] \equiv \varphi(q^{14})$ and

$$E[E[\varphi^7]\varphi^8] \equiv \varphi(q^7)E[\varphi^8] \equiv \varphi(q^7)(X^2 - XY + Y^2) \qquad (\text{as } 8 \equiv 1).$$

Also $1/\varphi(-q)^{16} \equiv \varphi(-q)^5/\varphi(-q^7)^3$, and iterating (b),

$$\frac{1}{\varphi(-q^7)^3} = \frac{\varphi(q^7)^3\varphi(q^{14})^6\varphi(q^{28})^{12}}{\varphi(-q^{56})^{24}}.$$

Multiplying numerator and denominator by $\varphi(q^{56})$:

$$S_{16}(q) \equiv \frac{\widetilde{H}(q)}{G(q^{56})} \pmod 7, \tag{1}$$

where

$$\widetilde{H} = (X^2 - XY + Y^2)\,\varphi(-q)^5\varphi(q^7)^4\varphi(q^{14})^6\varphi(q^{28})^{12}\varphi(q^{56}), \qquad G = \varphi(-q)^{24}\varphi(q).$$

The extra factor $\varphi(q^{56})$ makes the numerator's weight integral while preserving the denominator's support on multiples of 56 and its constant term 1. Write $\widetilde{H} = \sum K(m)q^m$. Since $1/G(q^{56})$ is a series in q^{56}, $\overline{p}(16n) \equiv \sum_{J=0}^{\lfloor n/56 \rfloor} e_J K(n - 56J)$ with $n - 56J \equiv n \pmod{56}$. Hence Theorem 1 follows from:

Proposition 4. *$K(m) \equiv 0 \pmod 7$ for all $m \equiv 3 \pmod 8$ with $\left(\frac{m}{7}\right) = 1$.*

4 Modularity and the Sturm bound

With $\varphi(q^d) = \eta(2d\tau)^5/(\eta(d\tau)^2\eta(4d\tau)^2)$ and $\varphi(-q^d) = \eta(d\tau)^2/\eta(2d\tau)$, each of the three monomials X^2Z, XYZ, Y^2Z of $\widetilde{H}$ (with Z the product of the remaining theta factors) is an eta-quotient of weight 18 and level 224. By the Gordon–Hughes–Newman criterion and Ligozat's cusp-order formula [2, Thms. 1.64, 1.65], each lies in $M_{18}(\Gamma_0(224), \chi)$ with $\chi = \left(\frac{14}{\cdot}\right)$ (we checked $\sum \delta r_\delta \equiv \sum (224/\delta) r_\delta \equiv 0 \pmod{24}$, the character, and nonnegative order at every cusp in exact arithmetic). Hence $\widetilde{H} \in M_{18}(\Gamma_0(224), \chi) \cap \mathbb{Z}[[q]]$.

Lemma 5 (Twisting at a sufficient level)**.** *Let $f = \sum a(m)q^m \in M_k(\Gamma_0(N), \chi)$, $k \in \mathbb{Z}$, let C be the conductor of χ, and let ψ be any Dirichlet character modulo M. Then, for $L = \operatorname{lcm}(N, M^2, MC)$,*

$$f \otimes \psi = \sum \psi(m)a(m)q^m \in M_k(\Gamma_0(L), \chi\psi^2).$$

No minimality of L or novelty of this twisting principle is asserted.

Proof. With $T_j = \left(\begin{smallmatrix} 1 & j/M \\ 0 & 1 \end{smallmatrix}\right)$ and $c_j = M^{-1}\sum_{r \bmod M} \psi(r)\zeta_M^{-rj}$ we have $f \otimes \psi = \sum_{j \bmod M} c_j f|T_j$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(L)$, multiplication gives

$$T_j\gamma T_{-d^2j} = \begin{pmatrix} a + cj/M & b - bcdj/M - cd^2j^2/M^2 \\ c & d - cd^2j/M \end{pmatrix}.$$

Its entries are integral since $M^2 \mid c$, its lower-left entry is divisible by N, and its lower-right entry is d modulo C since $MC \mid c$. Thus $(f|T_j)|\gamma = \chi(d)f|T_{d^2j}$. Reindexing gives $c_{d^{-2}j} = \psi(d)^2 c_j$, proving the law, even for imprimitive ψ. For any rational matrix α of positive determinant, choose $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ with $\sigma\infty = \alpha\infty$. Then $\sigma^{-1}\alpha$ is upper triangular with positive dilation ratio; it preserves nonnegative exponents in a holomorphic cusp expansion. Apply this to each translated cusp matrix $\alpha = T_j\gamma$. □

The indicator of $\{m \equiv 3\ (8),\ (\frac{m}{7}) = 1\}$ is a linear combination of real characters modulo 56:

$$\frac{1}{4}\sum_{\psi \bmod 8} \psi(3)\psi(m) \cdot \frac{1}{2}\big(\chi_0^{(7)}(m) + (\tfrac{m}{7})\big).$$

Here $C = 56$ and $L = \operatorname{lcm}(224, 56^2, 56^2) = 3136$. All twisting characters square to the principal character on units, so

$$g := \sum_{m \equiv 3(8),\, (m/7)=1} K(m)q^m \in M_{18}(\Gamma_0(3136), \chi) \cap \mathbb{Z}[[q]].$$

By Sturm's theorem with character [4, 2], the sufficient inclusive bound is

$$B = \frac{18}{12} \cdot 3136 \cdot \frac{3}{2} \cdot \frac{8}{7} = 8064.$$

Proposition 6 (computation)**.** *For all $m \le 8064$ with $m \equiv 3 \pmod 8$ and $\left(\frac{m}{7}\right) = 1$ (there are 432 such m), $K(m) \equiv 0 \pmod 7$. There are 370 nonzero complementary controls and $K(0) = 1$.*

This proves Proposition 4 and hence Theorem 1.

5 Computations

All proof checks are exact. From the collection root run `python verify.py mod7`. The identity checker compares the dissections with a separate overpartition recurrence; the modularity checker verifies the eta conditions; and both FLINT and a standard-library implementation compute the inclusive bound 8064, obtaining the 432 zero coefficients and 370 nonzero complementary controls of Proposition 6. The original larger-bound calculation is historical corroboration, not required by this proof. These are reproducible computational certificates, not formal proof-assistant verification.

For a compact modularity audit, the exponent vectors of X^2Z, XYZ, Y^2Z, in the order $\delta = (1,2,4,7,14,28,56,112,224)$, are

$$(-6,35,-16,-8,8,-2,46,-19,-2), \quad (10,11,-8,-8,8,-2,46,-19,-2),$$

$$(26,-13,0,-8,8,-2,46,-19,-2).$$

Their orders at cusp denominators $(1,2,4,7,8,14,16,28,32,56,112,224)$ are respectively

$$(70,120,6,10,12,72,2,42,0,84,14,0),$$

$$(126,64,6,18,12,64,2,42,0,84,14,0),$$

$$(182,8,6,26,12,56,2,42,0,84,14,0).$$

These are width-normalised cusp orders, so their nonnegativity proves holomorphy at every cusp.

Collection and verification

This paper is part of the revised collection, version 1.1.0-candidate, DOI `https://doi.org/10.5281/zenodo.23319157`. Companion papers and exact verification code are archived together. The statements are unrefereed computer-assisted proof candidates; no formal verification or unaffiliated reproduction is asserted.

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Overpartition congruences modulo 17, 19 and 23: proofs of three conjectures of Wei

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Abstract

Wei recently proved Ramanujan-type congruences for the overpartition function $\overline{p}(n)$ modulo 11 and 13, and stated three further congruences. Modulo 17 and 23 they were expected to follow from his method after Sturm-bound computations involving more than 10^9 coefficients, which were not completed. Modulo 19 the congruence was stated to be beyond the methods of the paper. We prove all three. For the moduli 17 and 23 we compute the Hecke operator $T(\ell^2)$ on a space of dimension at most 6 and isolate a correction term that vanishes on the relevant residue class; the computation needs fewer than 4000 coefficients. For the modulus 19 the relevant generating function is only weakly holomorphic. We multiply it by two eta-quotients congruent to 1 modulo 19 before projecting onto the required classes, and show that the projection is a cusp form of weight $35/2$ on $\Gamma_0(23104)$. The congruence then follows from a $T(17^2)$ eigen-relation checked up to the Sturm bound.

1 Introduction

Let $\overline{p}(n)$ denote the number of overpartitions of n [2], so that $\sum_{n\geq 0}\overline{p}(n)q^n = 1/\varphi(-q)$ with $\varphi(q)=\sum_{k\in\mathbb{Z}} q^{k^2}$. Wei [8] proved infinite families of congruences for $\overline{p}$ modulo 11 and 13 by Hecke operators on the spaces $M_k(\Gamma_0(4))$, which are spanned by products of φ and $F(q)=\sum_{n \text{ odd}}\sigma(n)q^n$. In his closing section he recorded the following.

Theorem 1. *For all $n\geq 0$,*

$$\overline{p}\big(17\cdot 11^2(88n+m)\big)\equiv 0 \pmod{17}, \qquad m\in\{19,35,43,51,83\},$$
$$\overline{p}\big(23\cdot 13^2(104n+l)\big)\equiv 0 \pmod{23}, \qquad l\in\{29,53,61,69,77,101\}.$$

Theorem 2. $\overline{p}(17^2\cdot 4n)\equiv 0 \pmod{19}$ *for every integer $n\geq 0$ with $n\equiv 3 \pmod 8$, $\left(\frac{n}{17}\right)=-1$ and $\left(\frac{n}{19}\right)=1$.*

Wei states the congruences of Theorem 1, explains that they reduce to Sturm-bound checks of about $1.2\cdot 10^9$ and $4.0\cdot 10^9$ coefficients on $\Gamma_1(340736)$ and $\Gamma_1(562432)$, and notes that these computations had not been completed. He states Theorem 2 as a conjecture that cannot be proved by the methods of [8]. A further conjecture of [8], modulo 7, is treated in the modulo-7 paper in this versioned collection.

Contribution. For Theorem 1, Wei had already reduced the statements to finite computations and expected them to follow from his method. Our contribution there is a short structural argument replacing a computation of more than 10^9 coefficients by one with fewer than 4000. Theorem 2 settles a conjecture that [8] places beyond its methods.

Relation to general results. Treneer's pole-removal method [7] is a standard input. Ryan, Sirolli, Villegas-Morales and Zheng [6, Theorems 1.3 and 1.5] already give square-prime, Legendre-restricted congruences as well as cube-prime families. A cube-versus-square distinction would therefore be incorrect. The comparison must use the actual parameters: our modulo-19 target arguments are prime to 19, whereas their displayed modulo-19 families contain that factor; the modulo-17/23 targets have the specified small primes and additional classes modulo 8. Wei already supplies their starting forms in his closing section. The remaining contribution is the stated proof and computational reduction, not the invention of those forms or of square-prime congruences. The consulted sources did not supply these exact conclusions; this is a bounded priority assessment.

The residue classes in Theorem 1 are exactly the N with $N \equiv 3 \pmod 8$ and $\left(\frac{N}{11}\right) = -1$, respectively $N \equiv 5 \pmod 8$ and $\left(\frac{N}{13}\right) = 1$. This was checked by enumeration modulo 88 and 104.

Method. Both proofs use Shimura's operators $T(\ell^2)$ on half-integral weight forms [4, 3]. For $f = \sum a(n)q^n \in M_{\lambda+1/2}(\Gamma_0(N))$ (trivial character, $4 \mid N$, $\ell \nmid N$),

$$(f \mid T(\ell^2))(n) = a(\ell^2 n) + \left(\frac{(-1)^\lambda n}{\ell}\right) \ell^{\lambda-1} a(n) + \ell^{2\lambda-1} a(n/\ell^2), \tag{1}$$

with $a(x) = 0$ for $x \notin \mathbb{Z}$. Theorem 1 needs only the low-dimensional spaces $M_{\lambda+1/2}(\Gamma_0(4))$. Theorem 2 concerns $\overline{p}(4n)$, whose generating function $\varphi(q)^3/\varphi(-q)^4$ has a pole at the cusp 0. The new ingredient is a holomorphic representative modulo 19 of the relevant projection.

Sturm's theorem. If $g \in M_k(\Gamma_0(N), \chi)$ has p-integral coefficients and $a(n) \equiv 0 \pmod p$ for $n \le \frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, then $g \equiv 0 \pmod p$ [5]. For half-integral weight we apply it to g^2 (or $g\varphi$), which has integral weight. If the first nonzero coefficient of g modulo p is at n_0, then that of g^2 is at $2n_0$, because $\mathbb{F}_p[[q]]$ is an integral domain.

2 Proof of Theorem 1

Write $(p, \ell, \lambda, r) = (17, 11, 7, 3)$ or $(23, 13, 10, 5)$.

Lemma 3 (Wei's reduction)**.** *There is $c = \sum c(n)q^n \in M_{\lambda+1/2}(\Gamma_0(4))$ with integer coefficients and $\sum \overline{p}(pn)(-q)^n \equiv c \pmod p$. Explicitly, with F as above,*

$$c = 13F^2\varphi^7 + 13F\varphi^{11} + \varphi^{15} \qquad (p = 17),$$
$$c = 20F^5\varphi + 17F^4\varphi^5 + 14F^3\varphi^9 + 5F^2\varphi^{13} + 9F\varphi^{17} + \varphi^{21} \qquad (p = 23).$$

Proof. We have $\sum \overline{p}(n)(-q)^n = 1/\varphi(q)$ and $\varphi(q)^p \equiv \varphi(q^p) \pmod p$, so $\varphi(q^p)\sum \overline{p}(n)(-q)^n \equiv \varphi^{p-1}$. Extracting the exponents divisible by p and replacing q^p by q gives $\varphi(q)\sum \overline{p}(pn)(-q)^n \equiv \varphi^{p-1} \mid T(p) \pmod p$, where $T(p)$ is the Hecke operator on $M_{(p-1)/2}(\Gamma_0(4), \chi_{-4}^{(p-1)/2})$ (its second term carries $p^{(p-3)/2} \equiv 0$). We computed $\varphi^{p-1} \mid T(p)$ exactly in the triangular basis $F^a\varphi^{p-1-4a}$. Its coordinates modulo p are $(1, 13, 13, 0, 0)$ for $p = 17$ and $(1, 9, 5, 14, 17, 20)$ for $p = 23$. Each is φ times the stated c. For $p = 17$ the coordinate of F^4 is 0, so no φ^{-1} term arises. Since $\varphi \in 1 + 2q\mathbb{Z}[[q]]$ is a unit modulo p, the claim follows. □

Proposition 4. *Modulo p, $c \mid T(\ell^2) \equiv \ell^{\lambda-1}c + h$ with $\ell^{\lambda-1} \equiv 8 \pmod{17}$, respectively $3 \pmod{23}$, and*

$$h = 12(F\varphi^{11} + F^2\varphi^7) \qquad (p = 17),$$
$$h = 16F\varphi^{17} + 16F^2\varphi^{13} + 14F^3\varphi^9 + 7F^4\varphi^5 + 15F^5\varphi \qquad (p = 23).$$

Proof. $T(\ell^2)$ preserves $M_{\lambda+1/2}(\Gamma_0(4))$, which has the upper-triangular basis $F^a\varphi^{2\lambda+1-4a}$ of dimension 4, respectively 6. We computed the integer matrix of $T(\ell^2)$ in this basis from (1), checking membership of each image in the span on all available coefficients. We then reduced $c \mid T(\ell^2) - \ell^{\lambda-1}c$ modulo p. □

Lemma 5. *$h(n) \equiv 0 \pmod p$ for all $n \equiv r \pmod 8$.*

Proof. Let $g = \sum_{n\equiv r\ (8)} h(n)q^n$. Since r is odd, averaging the twists of h by the four real characters modulo 8 shows $g \in M_{\lambda+1/2}(\Gamma_0(256))$ [8, Prop. 2.4 and Remark 2.5]. Then $g\varphi \in M_{\lambda+1}(\Gamma_0(256), \chi_{-4}^{\lambda+1})$ has integer coefficients. The Sturm bound is $(\lambda+1)\cdot 384/12 = 256$, respectively 352. The coefficients of $g\varphi$ vanish modulo p for $n < 4000$, so $g\varphi \equiv 0$, and $g \equiv 0$ because φ is a unit. □

Proof of Theorem 1. Let $N \equiv r \pmod 8$ with $\left(\frac{N}{\ell}\right) = -1$ (for $p = 17$), respectively $+1$ (for $p = 23$). Then $\ell \nmid N$, and the character value in (1) is $\left(\frac{-1}{11}\right)\left(\frac{N}{11}\right) = (-1)(-1) = 1$, respectively $\left(\frac{N}{13}\right) = 1$ since $\lambda = 10$ is even. Comparing N-th coefficients in Proposition 4,

$$c(\ell^2 N) + \ell^{\lambda-1}c(N) \equiv \ell^{\lambda-1}c(N) + h(N) \pmod p,$$

so $c(\ell^2 N) \equiv h(N) \equiv 0$ by Lemma 5. By Lemma 3, $\overline{p}(p\ell^2 N) \equiv (-1)^{\ell^2 N}c(\ell^2 N) \equiv 0 \pmod p$. □

3 Proof of Theorem 2

Let $c(n) = \overline{p}(4n)$. A 2-dissection gives

$$F_4 := \sum_{n\geq 0} c(n)q^n = \frac{\varphi(q)^3}{\varphi(-q)^4} = \frac{\eta(2z)^{19}}{\eta(z)^{14}\eta(4z)^6},$$

The identity follows algebraically, not from a finite check: with $E(\sum a_n q^n) = \sum a_{2n}q^n$, the classical relations $\varphi(q)\varphi(-q) = \varphi(-q^2)^2$, $E\varphi = \varphi(q^2)$ and $E(\varphi^2) = \varphi^2$ give successively

$$E(1/\varphi(-q)) = \varphi(q^2)/\varphi(-q)^2, \qquad E(\varphi(q^2)\varphi(q)^2/\varphi(-q^2)^4) = F_4.$$

Since $\varphi(q)$ and $\varphi(-q) = 2\varphi(q^4) - \varphi(q)$ transform under $\Gamma_0(16)$ with Shimura's theta multiplier $j(\gamma, z)$, F_4 is weakly holomorphic of weight $-1/2$. Its raw exponent at odd-denominator cusps is $-1/4$ (multiply by cusp width for the local-parameter order). Theorem 2 asserts $c(289n) \equiv 0 \pmod{19}$ on the stated classes.

A representative without poles at most cusps. Put $E_1 = \eta(z)^{19}/\eta(19z)$ and $E_2 = \eta(19z)^{19}/\eta(361z)$. Both lie in $1 + 19q\mathbb{Z}[[q]]$, because $(1-q^n)^{19} \equiv 1 - q^{19n}$. By the criterion of Gordon–Hughes–Newman and Ligozat, $E_1E_2 \in M_{18}(\Gamma_0(361))$ with trivial character. Set

$$F'' = F_4E_1E_2 = \frac{\eta(z)^5\eta(2z)^{19}\eta(19z)^{18}}{\eta(4z)^6\eta(361z)} \equiv F_4 \pmod{19}.$$

Then $F''(\gamma z) = j(\gamma, z)^{35}F''(z)$ on $\Gamma_0(16\cdot 361)$. Indeed $F_4(\gamma z) = j(\gamma, z)^{-1}F_4(z)$, and $(E_1E_2)(\gamma z) = (cz+d)^{18}(E_1E_2)(z)$. Since $j(\gamma, z)^2 = \left(\frac{-1}{d}\right)(cz+d)$, we have $(cz+d)^{18} = j(\gamma, z)^{36}$. So F'' is weakly holomorphic of weight $35/2$ with trivial character. Multiplying by an eta-quotient congruent to 1 modulo p to remove poles is the standard device of Treneer [7]. The point here is that the multiplication must precede the projection. At a cusp a/c with $361 \nmid c$, the factor E_1 (if $19 \nmid c$) or E_2 (if $19 \parallel c$) has order large enough to cancel the pole of F_4. So F'' has positive order at every cusp with $361 \nmid c$.

Projection. Let $G'' = \sum c''(n)q^n$, summed over $n \equiv 3 \pmod 8$ with $\left(\frac{n}{19}\right) = 1$, where $c''(n)$ are the coefficients of F''. Then $G'' \in \mathbb{Z}[[q]]$ and $G'' \equiv \sum_{n\equiv 3\,(8),\,(n/19)=1} c(n)q^n \pmod{19}$. Explicitly $G''(z) = \sum_{t \bmod 152} W_t F''(z + t/152)$. Writing $t = 19j + 8k$, the weights are $W_t = \frac{1}{8}e(-3j/8)\,b_k$ with $b_0 = 9/19$ and $b_k = -(1 + \chi(k)g)/38$, where $\chi = \left(\frac{\cdot}{19}\right)$ and g is the Gauss sum. These weights reproduce the indicator function exactly.

Lemma 6. *$G''(\gamma z) = j(\gamma, z)^{35} G''(z)$ for all $\gamma \in \Gamma_0(23104)$, where $23104 = 64 \cdot 361$.*

Proof. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(23104)$, so $152^2 \mid c$. For each t put $t' \equiv td^2 \pmod{152}$. Then $\gamma' = T_{t/152}\gamma T_{-t'/152}$ is in $\Gamma_0(16 \cdot 361)$ with lower-right entry $d' = d - ct'/152 \equiv d$ modulo 8 and modulo $c/23104$. Hence $j(\gamma', z + t'/152) = j(\gamma, z)$. Since $d^2 \equiv 1 \pmod 8$ and $\chi(kd^2) = \chi(k)$, we have $W_{t'} = W_t$, and the sum is permuted. □

Proposition 7. *G'' is a cusp form: $G'' \in S_{35/2}(\Gamma_0(23104))$.*

Proof. At the 228 cusps with $361 \nmid c$, every translate is expanded at a cusp whose reduced denominator is again not divisible by 361; exact order arithmetic gives strictly positive order in all $228 \cdot 152 = 34656$ cases. At the other twelve cusps, some translated summands have nonpositive exponents; this does not say that F'' itself has a pole at each of them. Appendix A specifies the complete finite obligation and expansion convention. Grouping by exponent and rational square class leaves 28 cyclotomic sums with 32928 summands. Every sum reduces exactly to zero modulo its cyclotomic polynomial. Thus all principal parts and constant terms vanish. This exact calculation, not numerical decay, proves the assertion.

Historical numerical corroboration used a separately written program; it is not the exact proof or a newly repeated scan. It evaluated $|G''(\gamma z)(cz + d)^{-35/2}|$ at 200-digit precision at all 240 cusps, along vertical rays of increasing height. It observed exponential decay everywhere, with stable fitted orders of 16, 19, 4 and 11 in the local parameter, and found no constant term or pole. At the 12 critical cusps the cancellation is by factors 10^{-63} to 10^{-118}, well within the working precision. The same program checked the transformation law on several matrices of $\Gamma_0(23104)$. □

Proof of Theorem 2. Since $17 \nmid 23104$, $T(289)$ acts on $S_{35/2}(\Gamma_0(23104))$ by (1) with $\lambda = 17$. Moreover $17^{16} \equiv 5$ and $17^{33} \equiv 17^{15} \pmod{19}$. Let $H = G'' \mid T(289) - 14G''$, a cusp form with integer coefficients. Its square belongs to $S_{35}(\Gamma_0(23104), \chi_{-4})$. The index is 36480, so the character-version Sturm bound is 106400. It suffices to check $H(n) \equiv 0 \pmod{19}$ for $0 \le n \le 53200$, inclusively. One implementation generates F_4 through exponent $289 \cdot 53200 = 15374800$ (with ten spare coefficients); a separate implementation generates $\overline{p}$ through argument $4 \cdot 289 \cdot 53200 = 61499200$. Both test the same finite obligation, with no exception. Hence, for all n in the support,

$$c(289n) + 5\left(\frac{n}{17}\right)c(n) + 17^{15}c(n/289) \equiv 14\,c(n) \pmod{19}.$$

If $n \equiv 3 \pmod 8$, $\left(\frac{n}{19}\right) = 1$ and $\left(\frac{n}{17}\right) = -1$, then $17 \nmid n$, $289n$ lies in the same classes, and the relation gives $c(289n) \equiv 19c(n) \equiv 0$. □

Remark 8. The scalar 14 differs from $1 + 17^{33} \equiv 8 \pmod{19}$; no particular Eisenstein series is being identified here. The relation does not extend to arbitrary coefficients outside the projection. Projecting F_4 before multiplying by E_1E_2 leaves poles, so the order of the two operations matters.

4 Computations

All computations use exact integer or modular arithmetic (python-flint), except the high-precision confirmation in Proposition 7. The scripts and logs are in the accompanying evidence bundle. Each theorem was also confirmed directly: Theorem 1 for arguments up to $2 \cdot 10^7$ (552 and 295 cases), and Theorem 2 for all $n \le 53200$.

A Exact cusp certificate

Write $e(x) = \exp(2\pi i x)$. The eta-exponent vector, in increasing argument order, is

$$(\delta, r_\delta) = (1,5), (2,19), (4,-6), (19,18), (361,-1).$$

Represent the cusps by a/c with $c \mid L = 23104$, $\gcd(a,c) = 1$, and one numerator in each unit class modulo $\gcd(c, L/c)$. There are 240 such cusps. Choose $A_s = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $ad - bc = 1$. The expansion convention is

$$(-i(c\tau + d))^{-35/2} G''(A_s\tau) = \sum_E C_E e(E\tau).$$

The cusp width is $w = L/\gcd(c^2, L)$, so the exponent in the width-normalised local parameter is wE. Multiplication by a nonzero phase to change slash conventions does not change vanishing.

c	cusps	w	raw $E \le 0$	groups
361	1	64	$-1/4, 0$	3
722	1	16	$-1, 0$	3
1444	2	4	$-4, 0$	6
2888	4	1	$-16, 0$	12
5776	2	1	0	2
11552	1	1	0	1
23104	1	1	0	1
Total	12			28

The listed exponents are those arising from translated summands before cancellation. All resulting group sums are zero.

Matrix and multiplier convention. For each $t = 19j + 8k$, decompose the integer matrix

$$M_t = \begin{pmatrix} 152a + tc & 152b + td \\ 152c & 152d \end{pmatrix} = \gamma \begin{pmatrix} u_1 & u_2 \\ 0 & u_3 \end{pmatrix}, \qquad \gamma \in \mathrm{SL}_2(\mathbb{Z}),\ u_1, u_3 > 0.$$

This is obtained by the extended Euclidean algorithm on the first column, with u_1 its positive gcd. For each eta factor similarly decompose $\mathrm{diag}(\delta, 1)\gamma = \gamma_\delta V_\delta$. For a matrix $(a', b'; c', d')$ with $c' > 0$, use

$$\eta(\gamma_\delta z) = e\left(\frac{a' + d'}{24c'} - \frac{s(d', c')}{2}\right)[-i(c'z + d')]^{1/2}\eta(z),$$

where s is the rational Dedekind sum and the analytic square root is fixed on the upper half-plane. This is the convention in Kong–Teo [1, Eq. (1.2)] after absorbing the $e(-1/8)$ phase into the square root.

Finite truncation and exact arithmetic. The decompositions give a leading exponent E_0, a root-of-unity phase, a positive square root of a rational number, and factors

$$\prod_{\delta}\prod_{n\geq 1}(1-e(n\beta_\delta)q^{n\alpha_\delta})^{r_\delta}, \qquad \alpha_\delta > 0.$$

If $E_0 > 0$, no term is needed. Otherwise expand only increments at most $-E_0$. All increments are positive; consequently omitted factors and higher binomial terms cannot contribute to an exponent $E \leq 0$. For $r < 0$ use $\binom{r}{m} = (-1)^m\binom{-r+m-1}{m}$. All phases and amplitudes before square roots are rational and retained exactly.

Multiply by the specified projection weights W_t and group by E and rational square class. Within each group, choose the least common denominator n of phases and form $P(X) = \sum_j q_j X^{nr_j \bmod n} \in \mathbb{Q}[X]$. Check $P \bmod \Phi_n = 0$. The actual fields have $n = 38, 76, 152$. This separate vanishing of all square-class groups is sufficient; no assumption of their linear independence is required. The certificate contains 32928 summands in 28 groups. Projection denominators divisible by 19 cause no reduction problem: G'' is defined as retention of integer coefficients, and the Fourier weights are used only to prove its modularity and cusp behaviour in characteristic zero.

Reproduction. From the collection root run `python verify.py mod17-19-23`. The wrapper pins the lifted eta vector, checks all 34656 positive orders, requires exactly twelve critical cusps and all 28 zero groups, and performs the inclusive Hecke–Sturm check. It fails on an unmet obligation, not merely prints a Boolean. Numerical cusp decay is retained as corroboration only.

Proposition 9 (Sufficient lift–project criterion)**.** *Let f be weakly holomorphic with integral Fourier coefficients, and let $E \equiv 1 \pmod{p}$ be an integral eta product or quotient. Suppose a finite Fourier projection P retains specified integer coefficients, $P(Ef)$ has a proved modular transformation law at a finite-index level, and every nonpositive cusp coefficient vanishes by complete finite expansions as above. Then $P(Ef)$ is an integral cusp form congruent to $P(f)$ modulo p. Any proposed Hecke congruence for it follows from a valid Sturm check in the resulting weight, level and character.*

Proof. The transformation law and cusp expansions establish cusp membership. Coefficient retention commutes with reduction, so $P(Ef) \equiv P(f)$. Apply the relevant Hecke operator and Sturm theorem to their difference. □

This is a sufficient certificate pattern, not an automatic existence theorem for a suitable E, nor a claim that this general principle is new.

Collection and verification

This paper is part of the revised collection, version 1.1.0-candidate, DOI `https://doi.org/10.5281/zenodo.23319157`. Companion papers and exact verification code are archived together. The statements are unrefereed computer-assisted proof candidates; no formal verification or unaffiliated reproduction is asserted.

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Class-restricted ℓ^2-congruences for overpartitions modulo 13, 17 and 23

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Abstract

We prove several new families of Ramanujan-type congruences for the overpartition function $\overline{p}(n)$, for example

$$\overline{p}(17 \cdot 4^i \cdot 29^2 \cdot m) \equiv 0 \pmod{17} \quad \text{for all } i \geq 0,\ m \equiv 3 \pmod 8,\ \left(\frac{m}{29}\right) = -1.$$

The displayed ℓ^2-families of Ryan, Sirolli, Villegas-Morales and Zheng for these moduli have Legendre restrictions but no additional class restriction modulo 8. Here small primes ℓ occur with such a class restriction. A filtered finite search suggested the examples. The proofs use Shimura's operators on low-dimensional modular-form spaces and correction terms that are reductions of exact $U(4)$ eigenvectors. One support identity therefore proves every tower level. We also classify exactly the levels at which the whole class $13 \cdot 4^i(8n+7)$ vanishes modulo 13.

1 Introduction

Let $\overline{p}(n)$ count the overpartitions of n [2], so that $\sum \overline{p}(n)q^n = 1/\varphi(-q)$ with $\varphi(q) = \sum_{k\in\mathbb{Z}} q^{k^2}$. Treneer [8] showed that $\overline{p}$ satisfies congruences $\overline{p}(m\ell^3 n) \equiv 0 \pmod m$, and Ryan, Sirolli, Villegas-Morales and Zheng [5] (also Zheng [10]) made these explicit. RSVZ give families of two kinds: $\overline{p}(m\ell^3 n) \equiv 0 \pmod m$, and $\overline{p}(m\ell^2 n) \equiv 0 \pmod m$ for n prime to ℓ with a single Legendre condition. Both kinds are infinite in ℓ for $m \in \{3,5,7,11\}$. For $m \in \{13,17,19\}$ they list the primes ℓ found by a search up to about 5000; in the ℓ^2-families the smallest are $\ell = 431$ for $m = 13$ and $\ell = 167$ for $m = 17$. For $m = 23$ they remark on $\ell = 5303$ and 8783. They note that cusp forms make the case $m \geq 13$ harder. None of these families has a congruence condition modulo 8. Wei [9] proved $\overline{p}(11(8n+5)) \equiv 0 \pmod{11}$ and $\overline{p}(13 \cdot 2^6(8n+7)) \equiv 0 \pmod{13}$. He also conjectured congruences modulo 7, 17, 19 and 23 involving residue classes modulo 8 together with Legendre conditions. Those modulo 17 and 23 are the cases $i = 0$ of (I) (with $\ell = 11$) and of (II) below, and they are proved here. A different treatment, together with the modulo-19 case, is in the modulo-17/19/23 paper in this versioned collection.

The discovery scan used $p \in \{13,17,19,23,29,31\}$, $\ell \in \{3,5,7,11,13,17,19,23,29,31,37,41,43\}$ with $\ell \neq p$, $0 \leq j \leq 6$, odd classes modulo 8, and both nonzero Legendre signs. Arguments were strictly below 40000000. It retained a candidate only after at least forty matching instances and discarded classes already vanishing without the ℓ^2 factor. Thus it was a filtered search, not an exhaustive classification of all possible congruences even within the parameter box. The logs are historical discovery evidence; the proofs below do not extrapolate them.

Theorem 1. *For all integers $i \geq 0$ and $m \geq 1$ satisfying the conditions below:*

(I) $\overline{p}(17 \cdot 4^i \ell^2 m) \equiv 0 \pmod{17}$ *for* $\ell \in \{11, 29\}$, $m \equiv 3 \pmod 8$, $\left(\frac{m}{\ell}\right) = -1$;

(II) $\overline{p}(23 \cdot 4^i \cdot 13^2 m) \equiv 0 \pmod{23}$ *for* $m \equiv 5 \pmod 8$, $\left(\frac{m}{13}\right) = 1$;

(III) $\overline{p}(13 \cdot 4^i \cdot 7^2 m) \equiv 0 \pmod{13}$ *for* $m \equiv 7 \pmod 8$, $\left(\frac{m}{7}\right) = -1$.

Moreover:

(IV) $\overline{p}(52\,\ell^2 m) \equiv 0 \pmod{13}$ *for* $\ell \in \{5, 29\}$, $m \equiv 1 \pmod 4$, $\left(\frac{m}{\ell}\right) = -1$; *and* $\overline{p}(104 \cdot 5^2 m) \equiv 0 \pmod{13}$ *for all odd* m *with* $\left(\frac{m}{5}\right) = 1$.

The cases $i = 0$ of (I) with $\ell = 11$ and of (II) are Wei's conjectures and coincide with the modulo-17/23 results in the companion paper in this collection; they are not separate contributions. The case $i = 3$ of (III) follows from Wei's theorem since $49m \equiv m \pmod 8$. In fact Proposition 4 below shows that precisely $i \equiv 3 \pmod 4$ have this whole-class explanation. The consulted literature did not establish the remaining exact parameter families; this bounded search is not universal priority evidence. The additional class conditions and actual prime choices, rather than square-prime shape alone, distinguish the displayed families.

2 Tools

Let $F = \sum_{n \text{ odd}} \sigma(n) q^n = \eta(4z)^8/\eta(2z)^4 \in M_2(\Gamma_0(4))$. For $\lambda \geq 0$ the space $V_\lambda = M_{\lambda+1/2}(\Gamma_0(4))$ has the basis $B_a = F^a \varphi^{2\lambda+1-4a}$, $0 \leq a \leq \lfloor (2\lambda+1)/4 \rfloor$, with $B_a = q^a + O(q^{a+1})$ [4, Thm. 1.49]. We use:

1. Shimura's operator [6] $T(\ell^2)$ (ℓ an odd prime) preserves V_λ, and acts by
$$(f \mid T(\ell^2))(n) = a(\ell^2 n) + \left(\frac{(-1)^\lambda n}{\ell}\right) \ell^{\lambda-1} a(n) + \ell^{2\lambda-1} a(n/\ell^2).$$

2. $U(4)$ preserves V_λ. Indeed, by [4, Prop. 3.7], $U(2)$ maps $M_{\lambda+1/2}(\Gamma_0(8), \psi)$ to $M_{\lambda+1/2}(\Gamma_0(8), \psi\left(\frac{8}{\cdot}\right))$, so $U(4) = U(2)^2$ preserves $M_{\lambda+1/2}(\Gamma_0(8)) \supseteq V_\lambda$. For each basis element, the difference D between $B_a \mid U(4)$ and the element of V_λ agreeing with it in the first $\dim V_\lambda$ coefficients satisfies $D\varphi \in M_{\lambda+1}(\Gamma_0(8), \chi_{-4}^{\lambda+1})$. This space has Sturm bound $\lambda+1$, and we checked $D\varphi = 0$ well beyond it. (This is also a theorem of Kohnen [3].)

3. The projection of $f \in V_\lambda$ onto an *odd* class modulo 8 lies in $M_{\lambda+1/2}(\Gamma_0(256))$, by averaging the four real characters [9, Prop. 2.4 and Remark 2.5]. Multiplying by φ gives weight $\lambda+1$ and character $\chi_{-4}^{\lambda+1}$, with Sturm bound $32(\lambda+1)$. The unit φ may then be cancelled. This character averaging does not project onto an even class.

4. (Wei's reduction [9].) Since $\varphi^p \equiv \varphi(q^p) \pmod p$, we have $\varphi \cdot \sum \overline{p}(pn)(-q)^n \equiv \varphi^{p-1} \mid T(p) \pmod p$, where the right side lies in $M_{(p-1)/2}(\Gamma_0(4), \chi_{-4}^{(p-1)/2})$. Let c_p be the element of $V_{(p-3)/2}$ with $c_p\varphi \equiv \varphi^{p-1} \mid T(p) \pmod p$; this is verified up to the Sturm bound on $\Gamma_0(4)$. Since φ is a unit in $\mathbb{F}_p[[q]]$, we have $\overline{p}(pN) \equiv (-1)^N c_p(N) \pmod p$. In the basis (B_a), c_p has coordinates modulo p equal to $(1, 4, 1)$ for $p = 13$, $(1, 13, 13, 0)$ for $p = 17$ and $(1, 9, 5, 14, 17, 20)$ for $p = 23$. These were verified up to the relevant Sturm bounds on $\Gamma_0(4)$ and directly against $\overline{p}$.

3 The mechanism

Fix p, $\lambda = (p-3)/2$ and $c = c_p$. Let $\mathcal{N}$ be a set of positive integers prime to ℓ on which $\eta := \left(\frac{(-1)^\lambda N}{\ell}\right)$ is constant. Put $h := c \mid T(\ell^2) - \eta\,\ell^{\lambda-1}c$. Comparing coefficients gives

$$c(\ell^2 N) \equiv h(N) \pmod{p} \qquad (N \in \mathcal{N}). \tag{1}$$

So it suffices to show that h vanishes on $\mathcal{N}$. For $\mathcal{N} = \{4^i m : \ m \equiv r\ (8)\}$ we use $h(4^i m) = (h \mid U(4)^i)(m)$. The coordinates of $h \mid U(4)^i$ modulo p run through a *finite* orbit under the $U(4)$-matrix. It therefore suffices to check, for each orbit element, that its projection to the class r modulo 8 vanishes, which is a single Sturm check on $\Gamma_0(256)$.

Lemma 2 (Support propagation)**.** *Suppose $c \mid T(\ell^2) - \epsilon\ell^{\lambda-1}c \equiv sH \pmod{p}$, $H \mid U(4) = \alpha H$, and $H(m) = 0$ on an odd class $r \pmod 8$. Then $c(\ell^2 4^i m) \equiv 0 \pmod{p}$ for every $i \geq 0$ and $m \geq 1$ in that class with $\left(\frac{(-1)^\lambda m}{\ell}\right) = \epsilon \in \{1, -1\}$.*

Proof. The Legendre condition ensures $\ell \nmid m$ and is unchanged by multiplication by 4^i. The backward Hecke term is therefore zero and the middle term cancels. The remaining coefficient is $sH(4^i m) = s\alpha^i H(m) = 0$. □

This elementary reusable lemma packages the existing Hecke argument; it is not claimed as a new general theorem in modular-form theory.

4 Proof of Theorem 1

The correction terms below are reductions of the following exact forms, written in the basis B_a of their respective spaces:

p	λ	coordinates of H	α	$h \bmod p$	zero class
13	5	$(0, 1, -16)$	16	$12H$	7
17	7	$(0, 1, -16, 0)$	-64	$12H$	3
23	10	$(0, 8, -383, 4032, 1280, -8192)$	-512	$2H$	5

Here $H \mid U(4) = \alpha H$ over the integers. Identify $U(4)$ in the integral triangular basis, then check the displayed eigenvector identities exactly. Projection of H to the listed odd class, multiplied by φ, has all coefficients zero through bounds $192, 256, 352$, respectively. Sturm over characteristic zero proves the exact support identities (or apply it modulo every prime). Hence Lemma 2 proves (I)–(III). The former orbit periods are simply the orders of $16, -64, -512$ modulo the corresponding primes, namely $3, 4, 22$.

The first form is $\eta(z)^2\eta(2z)^7\eta(4z)^2$, already listed by Aydin and Inam [1, Table 1, no. 43]. The second is $\eta(2z)^{27}/(\eta(z)^6\eta(4z)^6)$; negative eta exponents here do not imply cusp poles. These identities are auxiliary descriptions, not claims of novel eigenforms. The explicit modular checks below also provide an alternative finite-orbit proof.

(I) and (II). The relevant data modulo p are as follows; full matrices are in the evidence bundle.

p	ℓ	η	h (coordinates)	$U(4)$-orbit of h
17	11	$+1$	$(0, 12, 12, 0)$	period 4: $(0, 12, 12, 0) \to (0, 14, 14, 0) \to (0, 5, 5, 0) \to (0, 3, 3, 0)$
17	29	-1	$(0, 12, 12, 0)$	the same orbit
23	13	$+1$	$(0, 16, 16, 14, 7, 15)$	period 22

For $p = 17$ the sign is $\eta = \left(\frac{-1}{\ell}\right)\left(\frac{m}{\ell}\right)$. Since $\left(\frac{-1}{11}\right) = -1$ and $\left(\frac{-1}{29}\right) = 1$, we get $\eta = +1$ and -1 respectively; that is, $c \mid T(29^2) \equiv -29^6 c + h$. For $p = 23$, $\lambda = 10$ is even and $\eta = \left(\frac{m}{13}\right) = 1$. For $N = 4^i m$ the character value does not depend on i, since $\left(\frac{4^i}{\ell}\right) = 1$. The orbit is purely periodic and contains h itself, so it covers every $i \geq 0$. For every orbit element, the projection to the class $r = 3$ (respectively 5) modulo 8, multiplied by φ, vanishes modulo p for $n < 4000$. This exceeds the Sturm bounds 256 and 352 on $\Gamma_0(256)$. By (1), $\overline{p}(p\ell^2 4^i m) \equiv \pm c(\ell^2 4^i m) \equiv \pm h(4^i m) \equiv 0$.

(III). Here $p = 13$, $\lambda = 5$, $\ell = 7$, and $\eta = \left(\frac{-4^i m}{7}\right) = \left(\frac{-1}{7}\right)\left(\frac{m}{7}\right) = 1$. Then $h = c \mid T(49) - 7^4 c$ has coordinates $(0, 12, 3)$. Its $U(4)$-orbit modulo 13 is the 3-cycle (containing h, so it covers all i) $(0, 12, 3) \to (0, 10, 9) \to (0, 4, 1)$. Each projection to the class 7 modulo 8, times φ, vanishes modulo 13 beyond the Sturm bound 192.

(IV). For $N = 4m$ with $m \equiv 1 \pmod 4$ and $\left(\frac{m}{\ell}\right) = -1$, $\ell \in \{5, 29\}$, we have $\eta = \left(\frac{-1}{\ell}\right)\left(\frac{m}{\ell}\right) = -1$. So $h_\ell = c \mid T(\ell^2) + \ell^4 c$ (each $T(\ell^2)$-image was identified in the 3-dimensional space V_5 from its first coefficients and checked well beyond), with coordinates $(7, 9, 12)$ for $\ell = 5$ and $(5, 12, 3)$ for $\ell = 29$. Then $h_\ell(4m) = (h_\ell \mid U(4))(m)$, and the projection of $h_\ell \mid U(4)$ to the classes $1, 5$ modulo 8 vanishes (Sturm bound 192). For $N = 8m$ with m odd and $\left(\frac{m}{5}\right) = 1$, $\eta = \left(\frac{-8m}{5}\right) = \left(\frac{-1}{5}\right)\left(\frac{2}{5}\right)\left(\frac{m}{5}\right) = -1$, so again $c(25N) \equiv h_5(N)$. With $k = h_5 \mid U(4)$ we need $k(2m) \equiv 0$ for odd m. The form $w = k \mid U(2) - (k \mid U(4)) \mid V(2)$ lies in $M_{11/2}(\Gamma_0(8), \left(\frac{8}{\cdot}\right))$, and its coefficients are $k(2m)$ for m odd and 0 for m even. Moreover $w\varphi$ vanishes modulo 13 beyond its Sturm bound 6. Since $\overline{p}(13N) \equiv \pm c(N)$, this proves (IV). □

Remark 3. The congruences need their specified conditions; the finite search is not a classification of all other classes or primes. The exact eigenvector identities explain propagation without testing every orbit state. Other primes can be tested by the same finite-space calculation, but adding a list of numerical cases alone would not strengthen the theorem.

Proposition 4 (Exact whole-class classification modulo 13)**.** *For every integer $i \geq 0$,*

$$\left[\overline{p}(13 \cdot 4^i(8n + 7)) \equiv 0 \pmod{13} \text{ for every } n \geq 0\right] \iff i \equiv 3 \pmod 4.$$

Proof. In the row-coordinate convention, the exact matrix and initial vector are

$$U = \begin{pmatrix} 1 & 5280 & 42240 \\ 0 & 336 & 2816 \\ 0 & 20 & 192 \end{pmatrix}, \qquad c_{13} = (1, 4, 1) \pmod{13}.$$

Its orbit modulo 13 has length 12 and returns to its initial vector. For each of its twelve vectors, projection onto class 7, followed by multiplication by φ, is tested through the inclusive Sturm bound 192. It vanishes precisely for states $3, 7, 11$. For every other state the coefficient at 7 is the nonzero witness

$$c_{13}(4^i 7) \equiv 7 + 4 \cdot 5^i \pmod{13}.$$

This formula is checked on all twelve states and is periodic, since 5 has order 4 modulo 13. The sign in $\overline{p}(13N) \equiv (-1)^N c_{13}(N)$ does not affect vanishing. Thus both directions hold for every i, not just the discovery range. □

The classification was suggested by the supplied reviewer audit and checked in the revised collection. Wei supplies the $i = 3$ instance; the extension's literature status is bounded by the recorded search.

5 Verification

All computations use exact or modular arithmetic (python-flint). They were carried out by two separate scripts (for $p \in \{17, 23\}$ and $p = 13$) and confirmed by a third, independent checker written by a separate reviewer. Each congruence was also checked directly from $\overline{p} \bmod p$: up to arguments $3 \cdot 10^6$ for (I) and (II), and up to $4 \cdot 10^7$ in the original search, with no exceptions.

Collection and verification

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